

MoCafe v2.00 — A Monte-Carlo Dust Radiative-Transfer Code with Scattering and Thermal Emission: User Guide

Abstract

MoCafe is a parallel (Fortran 90 + MPI) Monte-Carlo dust radiative-transfer code. Version 2.00 extends the scattering-only v1.20 into a panchromatic code that computes *dust thermal emission*: it transports photon packets over a wavelength grid, tallies the radiation field in every cell, and re-emits the absorbed energy as thermal dust emission through either the Lucy (1999) method coupled to the SEDUST grain library (equilibrium and stochastically heated grains, including PAH features) or the Bjorkman & Wood (2001) immediate-reemission method. It also adds multiple stellar populations, an all-sky interior observer (for the Milky-Way case), and a Modified Random Walk for very optically thick regions. This guide covers installation, execution, the complete input reference, the dust-emission workflow, the output products, worked examples, and the underlying algorithms.

Contents

1. Overview	3
2. Installation	3
2.1 Prerequisites	3
2.2 Building MoCafe	4
2.3 Building the SEDUST library	4
3. Running MoCafe	4
4. Input Parameter Reference	4
4.1 Simulation control	5
4.2 Grid geometry (Cartesian)	5
4.3 Grid type and media	5
4.4 Distance unit	6
4.5 Source geometry (single source)	6
4.6 Dust and scattering (grey / monochromatic)	6
4.7 Optical depth and density	6
4.8 Peel-off observers	7
4.9 Output control	7
5. Dust Thermal Emission	7
5.1 Multi-wavelength (SED) mode	7
5.2 Mean intensity J_λ in each cell	8
5.3 Mode 1: Lucy (1999) with SEDUST	8
5.4 Mode 2: Bjorkman & Wood (2001)	9

5.5	Multiple stellar populations	10
5.6	HEALPix all-sky interior observer (Milky Way)	10
5.7	Modified Random Walk (high optical depth)	10
5.8	Dust emission on the AMR octree	11
5.9	A minimal emission run	11
6.	Output Products	12
7.	Worked Examples	12
8.	Validation	13
9.	Algorithm Notes	13
	References	14

1. Overview

MoCafe follows photon packets emitted by one or more sources through a dusty medium. Packets scatter off dust grains (Heney–Greenstein phase function or a tabulated Mueller matrix) and, in v2.00, are absorbed and re-emitted as thermal radiation. At every interaction a contribution is “peeled off” toward each observer to build an image or spectral-energy distribution (SED).

What is new in v2.00.

- **Multi-wavelength (SED) transport** (`par%use_sed`): one run covers the full UV-to-far-IR range on a log-spaced wavelength grid.
- **Mean intensity in each cell** $J_\lambda(x, y, z)$ (`par%save_jlam`) via the Lucy (1999) pathlength estimator.
- **Dust thermal emission** (`par%use_dustemis`) in two modes: *Mode 1* (Lucy + SEDUST) and *Mode 2* (Bjorkman & Wood 2001).
- **Multiple stellar populations** (`par%nsources`) with independent spectra, luminosities, and spatial distributions.
- **All-sky interior observer** (`par%allsky`) for the Milky-Way case.
- **Modified Random Walk** (`par%use_mrw`) plus Russian roulette for very optically thick media.

Everything above is opt-in; a plain monochromatic scattering run behaves exactly as in v1.20. The design and validation record is in `MoCafe_v2.00_PLAN.md`.

Units. In scattering-only mode MoCafe is unit-agnostic and the image is in (luminosity unit) $\text{cm}^{-2} \text{sr}^{-1}$. In SED / emission mode a physical length scale is required (through `par%distance_unit`: one grid length unit equals one distance unit) and the source luminosity (`par%luminosity`) is in erg s^{-1} , so intensities and dust temperatures are absolute.

2. Installation

2.1 Prerequisites

- A Fortran 2008 MPI compiler: Intel oneAPI (`mpiifort` or `mpiifx`) or GNU (`mpif90`). MoCafe is MPI-only.
- **CFITSIO** (required).
- **HDF5** 1.14+ with Fortran bindings built by the same compiler (optional; on by default).
- Python 3 with numpy, astropy, h5py (and scipy for the Illustris converter) for the analysis tools.

2.2 Building MoCafe

```
make                # HDF5=1 (default) -> MoCafe.x
make HDF5=0         # CFITSIO only
make HDF5_PREFIX=/usr/local/hdf5 # override HDF5 location
make F90=mpif90     # force a compiler
make DEBUG=1        # checked / traceback flags
```

The compiler is auto-selected `mpiifort` → `mpiifx` → `mpif90`. The build always passes `-cpp` `-DMPI` and (by default) `-DHDF5`. When you add a source file, append its object to `OBJs` in the Makefile in dependency order.

2.3 Building the SEDUST library

The Lucy emission engine (Mode 1) links the SEDUST library, kept self-contained under `SEDust/`. Build it once:

```
cd SEDust/sed && ./build_lib.sh # -> SEDust/sed/lib/libsedust.a (Intel)
```

The MoCafe Makefile links `SEDust/sed/lib/libsedust.a` automatically (variable `SEDUST_LIBDIR`). The SEDUST module named `mathlib` is renamed to `sed_mathlib` to avoid a clash with MoCafe's own `mathlib`. The optics tables the default `astrodust` path reads (~26 MB) are committed to the repository, and `par%sed_workdir` defaults to blank so MoCafe auto-resolves the SEDUST directory relative to the `MoCafe.x` executable at run time — a fresh checkout runs dust emission from any working directory with no path editing. Only the 17 MB D16 spheroid Q-library (used by a non-default graphite path) is kept out of git; run `SEDust/populate_data.sh` to fetch it, editing its source path (or setting `SEDUST_SRC`) to your own SEDUST tree. Mode 2 (Bjorkman & Wood) needs only the extinction table and not SEDUST.

3. Running MoCafe

```
mpirun -np <N> ./MoCafe.x <input>.in
```

The single argument is a Fortran namelist file. Every tunable quantity is a field of the global `par` structure and is set as `par%<name> = <value>` inside a `¶meters / . . . / block`. Sample inputs and `run.sh` scripts live under `examples/`. The default of every parameter is the initializer of `params_type` in `src/define.f90`.

Reproducibility. A run is bit-identical when `par%iseed` is a fixed non-zero value, `par%use_master_slave` is `.false.` (equal-share), and the number of MPI ranks is fixed (`par%iseed = 0` seeds from the clock/PID).

4. Input Parameter Reference

Parameters are grouped below. Types: L logical, I integer, R real, S string, [] array.

4.1 Simulation control

Parameter	Default	Description
no_photons	1e5	Number of photon packets
no_print	1e7	Progress-print interval (photons)
iseed	0	RNG seed (0 $\rightarrow$ seeded from clock/PID)
use_master_slave	.true.	MPI mode: master-slave vs. equal-share
num_send_at_once	10000	Batch size (master-slave)
luminosity	1.0	Total source luminosity [erg/s] (SED/emission mode)

4.2 Grid geometry (Cartesian)

Parameter	Default	Description
nx, ny, nz	1,1,11	Grid resolution
xmax, ymax, zmax	1.0	Half-sizes of the box (grid spans $[-\text{max}, \text{max}]$)
rmax	-999	If > 0 and not a cylinder, forces a cube of side r_{max} (sphere)
nr	-999	If > 1 , sets $\text{nx}=\text{ny}=\text{nz}=\text{nr}$
geometry	''	'', 'sphere', 'cylinder', or 'Plummer' (auto-set from source)
xyz_symmetry	.false.	Exploit octant symmetry (disabled if any $n = 1$)
z_symmetry	.false.	Exploit reflection about $z = 0$
xy_periodic	.false.	Periodic in x, y

4.3 Grid type and media

Parameter	Default	Description
grid_type	'car'	'car' (Cartesian), 'clump' (clumpy), 'amr' (octree)
amr_file	''	Generic AMR octree file (FITS/HDF5/text) for 'amr'
dust_model	'global_dgr'	AMR leaf opacity: 'global_dgr', 'from_file', 'laursen09'
DGR, Z_global, Z_ref, f_ion_dust	—	Dust-to-gas / metallicity parameters (AMR)
clump_radius	-1	Clump radius (clumpy medium)
clump_f_cov / _f_vol / _N_clumps	-1	Clump count specification
clump_tau0 / _ndust / _nH	-1	Opacity specification for each clump

The clumpy and AMR media are documented in docs/MoCafe_clump.pdf and docs/MoCafe_amr.pdf. The full dust-emission suite runs on both the Cartesian grid and the AMR octree (§5.8); the clumpy medium is scattering-only.

4.4 Distance unit

Parameter	Default	Description
distance_unit	'kpc'	'', 'pc', 'au', 'kpc'; sets distance2cm. In SED mode one grid unit = one distance unit (physical scale)

4.5 Source geometry (single source)

Parameter	Default	Description
source_geometry	'point'	'point', 'uniform', 'uniform_xy', 'gaussian', 'exponential', 'external_{sph,cyl,rec}'
xs_point, ys_point, zs_point	0.0	Point-source location
source_zscale	NaN	Scale height for 'gaussian'/'exponential'
radiation_angular_PDF_file	''	Angular PDF for external illumination

Multiple sources are described in §5.5.

4.6 Dust and scattering (grey / monochromatic)

Parameter	Default	Description
albedo	0.3253	Single-scattering albedo
hgg	0.6761	Henye-Greenstein asymmetry g
cext_dust	1.61e-21	Dust extinction per H nucleus [cm ²]
scatt_mat_file	''	Mueller-matrix table (data/mueller_*.dat); enables Stokes
use_stokes	.true.	Stokes polarization (auto-off without a matrix file)

In SED mode the wavelength-dependent albedo, g , and C_{ext} are read from par%kext_file and override these scalars (§5.1).

4.7 Optical depth and density

Parameter	Default	Description
taumax	-999	Radial optical depth, center → edge (sets the density scale)
tauomo	-999	Optical depth of the volume-equivalent homogeneous medium
density_file	''	3-D density cube (FITS/HDF5)
reduce_factor	1	Density-file down-sampling

Parameter	Default	Description
centering	0	Density-file re-centering mode
density_zscale, density_rscale, density_powerlaw_index	NaN	Analytic density modulations

In SED mode `par%taumax/par%tau homo` are defined at the reference wavelength `par%lambda_ref`.

4.8 Peel-off observers

Parameter	Default	Description
nxim, nyim	129	Image pixels
dxim, dyim	NaN	Pixel size [deg] (auto to cover the grid if unset)
distance	NaN	Observer distance; renormalizes (obsx, obsy, obsz)
inclination_angle, position_angle, phase_angle	NaN	Observer angles [deg] (arrays)
alpha, beta, gamma	NaN	Euler angles [deg] (alternative)
obsx, obsy, obsz	NaN	Explicit observer direction (alternative)

Up to `MAX_OBSERVERS` = 181 observers per run. Angle conventions are in `docs/MoCafe_Geometry.pdf`.

4.9 Output control

Parameter	Default	Description
file_format	'hdf5'	'hdf5' or 'fits'; authoritative for the container
out_file	' '	Output base name (from the input name if empty)
out_bitpix	-32	FITS bitpix (−32 float32, −64 float64)
save_direct0	.false.	Also write the unattenuated direct image
sightline_tau	.false.	Also write a sight-line optical-depth map for each pixel

5. Dust Thermal Emission

The emission workflow is: *SED transport* (panchromatic scattering) $\rightarrow J_\lambda$ tally $\rightarrow$ *dust emission* $\rightarrow$ observer SEDs. Enable it with `par%use_sed`, and add `par%save_jlam` and `par%use_dustemis` for the emission.

5.1 Multi-wavelength (SED) mode

Parameter	Default	Description
use_sed	.false.	Enable panchromatic transport
nlambda	128	Number of log-spaced wavelength bins
lambda_min, lambda_max	0.0912, 2000	Wavelength range [μm]
lambda_ref	0.55	Reference wavelength [μm] for the grid opacity and τ
kext_file	''	Table λ , κ , $\langle\cos\rangle$, C_{ext}/H
source_spectrum	''	Two-column source spectrum λ , L_λ
tstar	-999	Planck source temperature [K] (if no spectrum file)

Wavelength-dependent extinction is applied as a grey rescaling for each photon

$$s_{\text{ext}}(\lambda) = \frac{C_{\text{ext}}(\lambda)}{C_{\text{ext}}(\lambda_{\text{ref}})}, \quad (1)$$

so the grid stores only the reference-wavelength opacity and every optical depth is multiplied by s_{ext} (the same device as the Jonsson 2006 τ scan). All three ray tracers (Cartesian, clumpy, AMR) are therefore shared with the monochromatic path, and a run with $s_{\text{ext}} = 1$ is bit-identical to the mono code.

Restrictions: SED mode is incompatible with Stokes polarization and with the $(a, g)/\tau$ single-run scans, and (in this version) supports internal sources only.

5.2 Mean intensity J_λ in each cell

Parameter	Default	Description
save_jlam	.false.	Tally J_λ and write <base>_jlam

The mean intensity is accumulated with the Lucy (1999) pathlength estimator: each path segment of length $d\ell$ contributes $wL_{\text{pk}}d\ell$ to its (wavelength-bin, cell) slot, and

$$J_\lambda(\text{cell}) = \frac{1}{4\pi V_{\text{cell}} \Delta\lambda d_{\text{cm}}^2} \sum_{\text{paths}} wL_{\text{pk}} d\ell, \quad (2)$$

where L_{pk} is the packet luminosity [erg/s] and d_{cm} the distance-unit conversion. The forced first-flight is tallied analytically (a zero-variance direct term). A built-in energy check reports the absorbed luminosity from the tally against an independent event counter. Memory is $\approx n_\lambda \times n_{\text{cell}} \times 8$ bytes per MPI rank ($n_{\text{cell}} = n_x n_y n_z$ on the Cartesian grid, or the number of leaves on the AMR octree; see §5.8).

5.3 Mode 1: Lucy (1999) with SEDUST

Parameter	Default	Description
use_dustemis	.false.	Compute dust emission (requires par%use_sed)

Parameter	Default	Description
dust_emission_method	'lucy'	'lucy' (SEDUST) or 'bw01' (Bjorkman & Wood)
dust_model_sed	'astrodust'	SEDUST grain model: 'astrodust', 'dl07', or 'zubko'
sed_qtable, sed_sizedist	SEDust/	optics / size-distribution paths ('astrodust', 'dl07')
sed_dl07_sdindex, sed_dl07_uisrf	7, 1.0	'dl07' only: WD01 size-dist. index and reference U
sed_zubko_config, sed_zubko_dir	../data/zubko/	'zubko' only: ZDA config + DustEM data dir
sed_workdir	blank (auto)	SEDUST data dir; blank auto-resolves to <MoCafe.x dir>/SEDust/sed
sed_NT, sed_Tlo, sed_Thi	200, 2.7, 5000	Grain temperature grid
dust_niter	1	Lucy iterations (dust self-absorption)
dust_no_photons	1e6	Dust-emission photons per iteration
dust_tol	1e-3	Convergence on the total emitted luminosity

The 'lucy' method requires `par%save_jlam`. For each cell the local mean intensity J_λ (converted to SI and resampled onto the SEDUST wavelength grid) is passed to SEDUST, which returns the emission spectrum from the full grain-size distribution — equilibrium grains plus stochastically heated small grains and PAH features. The *shape* is taken from SEDUST while the bolometric luminosity is set to the locally absorbed power,

$$L_{\text{emit}}(\text{cell}) = L_{\text{abs}}(\text{cell}) = \rho\kappa \sum_{\lambda} s_{\text{ext}}(\lambda) [1 - \mathfrak{W}(\lambda)] \left(\sum_{\text{paths}} w L_{\text{pk}} d\ell \right)_{\lambda, \text{cell}}, \quad (3)$$

so total energy is conserved by construction (radiative equilibrium). With `par%dust_niter` > 1 the re-emitted photons are transported so that dust self-absorption heats other cells; the iteration converges when the total emitted luminosity stops changing (needed at high infrared optical depth).

Emission-solver speed options. The exact stochastic solve costs ~ 0.1 s per cell; two faster paths are available:

Parameter	Default	Description
dust_single_teq	.false.	One mixture equilibrium T per cell; fastest ($\sim 55\times$); true T_{eq} ; <i>no PAH/stochastic features</i>
dust_fast_table	.false.	Interpolate emission vs. field intensity U over a fixed reference shape; $\sim 1\%$ SED-shape error; keeps stochastic/PAH
dust_nU	40	U -grid size for the table path

5.4 Mode 2: Bjorkman & Wood (2001)

Set `par%dust_emission_method` = 'bw01'. This is an approximate, equilibrium, mixture mean-opacity method with *no iteration*: fixed-energy packets random-walk through scatter and absorb+immediate-reemit events until they escape.

Each absorption raises the local temperature (from the running absorbed power) and the packet wavelength is resampled from the temperature-correction spectrum $\propto \kappa_{\text{abs}} \partial B_{\lambda} / \partial T$, so the time-averaged emission equals $\kappa_{\text{abs}} B_{\lambda}(T)$ without iterating. Mode 2 needs only `par%kext_file` (not SEDUST) and does not reproduce PAH or stochastic-heating features. It writes `<base>_bwdust` (equilibrium T_{dust} and absorbed-power maps); the emergent dust emission is added to the observer Scattered SED image.

5.5 Multiple stellar populations

Parameter	Default	Description
<code>nsource</code>	1	Number of stellar components (> 1 activates multi-source)
<code>src_geometry(i)</code>	'point'	Geometry of each source: point/uniform/gaussian/exponential
<code>src_tstar(i)</code>	-999	Planck temperature of each source [K]
<code>src_spectrum(i)</code>	' '	Spectrum file of each source (overrides <code>src_tstar</code>)
<code>src_lum(i)</code>	-999	Luminosity of each source [erg/s] (equal split if unset)
<code>src_x/y/z(i)</code>	0	Position of each source ('point')
<code>src_zscale(i)</code>	-999	Scale height of each source ('gaussian'/'exponential')

Sources are sampled in proportion to their luminosity, and `par%luminosity` becomes $\sum_i \text{src_lum}(i)$. A typical use is a hot young disk plus a cool old bulge (examples/galaxy/).

5.6 HEALPix all-sky interior observer (Milky Way)

Parameter	Default	Description
<code>allsky</code>	.false.	Observer inside the medium; bin every event by sky direction
<code>allsky_x/y/z</code>	0	Interior observer position (code units)
<code>allsky_nside</code>	64	HEALPix resolution (power of 2); $n_{\text{pix}} = 12n_{\text{side}}^2$

Each scatter, dust-reemission, and direct event peels toward the interior observer and is binned into the **HEALPix** pixel (RING scheme, equal-area pixels) of the sky direction it arrives from, following LaRT's `observer_heal` implementation. The event-to-observer optical depth is integrated only over the segment to the observer, using the Cartesian or the octree segment ray trace as appropriate — so the all-sky map works on **both the Cartesian grid and the AMR octree**. The output `<base>_allsky` is a HEALPix $(n_{\text{pix}}, \lambda)$ surface-brightness map — the diffuse Galactic light and far-infrared cirrus as seen from inside the disk. It carries the standard HEALPix keywords (PIXTYPE, ORDERING='RING', NSIDE, NPIX, OMEGAPIX) and a `PixelDir` table of pixel-center unit vectors, so it reads directly with `healpy` (examples/milkyway/).

5.7 Modified Random Walk (high optical depth)

Parameter	Default	Description
use_mrw	.false.	Diffusion step in scattering-thick cells
mrw_gamma	2.0	Trigger threshold on the nearest-wall scattering optical depth

In a cell whose nearest-wall *scattering* optical depth exceeds `par%mrw_gamma`, MoCafe replaces the many small steps with a single diffusion step to the inscribed-sphere surface (Min et al. 2009; Robitaille 2010): it samples the total path length inside the sphere from the diffusion first-passage distribution, deposits the absorption-weighted path into the J_λ tally, and decays the weight by $\exp(-\alpha_{\text{abs}} ct)$. The Lucy transport also applies Russian roulette so high-albedo, high- τ clouds do not diffuse indefinitely. MRW is exact in absorbing dust (where it triggers rarely); its speed-up is large only when individual cells are very thick ($\tau_{\text{cell}} \gg 1$) and nearly pure-scattering.

5.8 Dust emission on the AMR octree

The entire dust-emission suite — SED transport, the J_λ tally, Mode 1 (Lucy, including the single- T_{eq} and table options), Mode 2 (Bjorkman & Wood), and the Modified Random Walk — runs on the AMR octree (`par%grid_type = 'amr'`) exactly as on the Cartesian grid. A “cell” is simply an octree leaf: the radiation field, absorbed power, and dust temperature are tallied per leaf (code-volume and opacity of each leaf), and MRW uses the nearest leaf-face distance for its trigger. Set up the octree as usual (`par%grid_type = 'amr'`, `par%amr_file`, and `par%dust_model` for the leaf opacity) and add the emission parameters above. The derived files carry arrays indexed by leaf instead of Cartesian cubes: `<base>_jlam`, `<base>_dustsed`, and `<base>_bwdust` each add a LeafXYZ table of leaf-center coordinates so the leaf quantities can be placed in space. Example: `examples/amr_sphere/amr_dustemis.in` (a uniform level-5 octree sphere) reproduces the energy budget and SED of the equivalent Cartesian sphere.

5.9 A minimal emission run

```
&parameters
par%no_photons = 2.0e6
par%use_sed     = .true.
par%save_jlam   = .true.
par%use_dustemis = .true.
par%dust_single_teq = .true.      ! fast equilibrium solve
par%luminosity  = 3.828e33        ! erg/s
par%nlambda     = 100
par%lambda_min  = 0.0912
par%lambda_max  = 2000.0
par%kext_file   = 'data/kext_astrodust_MW.dat'
par%tstar       = 1.0e4
par%taumax      = 5.0
par%source_geometry = 'point'
par%distance_unit = 'pc'
```

```
par%rmax = 1.0
par%nx = 33
par%ny = 33
par%nz = 33
par%distance = 1.0e5
/
```

6. Output Products

par%file_format is authoritative: if par%out_file's extension disagrees, MoCafe rewrites it (with a warning) so that the main file and all derived files share one container. HDF5 mirrors the FITS HDU layout (each image HDU becomes a group named after its EXTNAME with a data dataset; FITS keywords become group attributes).

Suffix	Written when	Contents
_obs	always	Observer images; in SED mode Scattered/Direct become (x, y, λ) cubes plus the wavelength / optics tables
_jlam	save_jlam	$J_\lambda(\lambda, x, y, z)$, J_{bol} , energy-conservation keywords
_dustsed	use_dustemis (‘lucy’)	Emergent and intrinsic dust SED, a pixel-resolved DustEmis_image(x, y, λ), and $T_{\text{dust}}/L_{\text{dust}}$ maps for each cell
bwdust	‘bw01’	Equilibrium T{dust} and absorbed-power maps
allsky	allsky	HEALPix (n{pix}, λ) all-sky surface-brightness map (RING) + PixelDir
_obs_tau	sightline_tau	Sight-line optical-depth map
_stokes	use_stokes	Stokes I, Q, U, V images

On the AMR octree the _jlam, _dustsed, and _bwdust images are arrays indexed by leaf rather than Cartesian cubes, and each adds a LeafXYZ table of leaf-center coordinates. Read and convert the outputs with python/mocafe_io.py (info, peek, convert).

7. Worked Examples

Directory	Shows
examples/point_source, uniform_source, external_*	Cartesian grids and source geometries
examples/agtau_list	(a, g) and τ single-run scans
examples/clump_sphere, amr_sphere, amr_tng, amr_ramses	Clumpy and AMR media

Directory	Shows
examples/dustemis	Dust emission: Lucy+SEDUST, iteration, single- T_{eq} , table, B&W, MRW
examples/amr_sphere	Dust emission on the AMR octree (amr_dustemis.in; matches the Cartesian sphere)
examples/multipop	Two stellar populations (hot young + cool old)
examples/galaxy	External galaxy SED (disk + bulge, edge-on / face-on)
examples/milkyway	All-sky map from an interior observer
examples/benchmarks	Camps (2015) SHG dust-emission benchmark and Mode 1/2 τ sweep

8. Validation

- **Regression.** The monochromatic scattering, the $(a, g)/\tau$ scans, and the clumpy-medium outputs remain bit-identical to v1.20 throughout the upgrade.
- **Dust-emission engine.** Run through the SEDUST library with the Zubko et al. (2004) BARE-GR-S model, the emission spectra agree with the Camps et al. (2015) seven-code stochastic-heating benchmark to a median relative difference of 2–4% across $U = 10^{-2} \dots 10^4$, and the emission peak matches the code median to within one wavelength bin.
- **Energy conservation.** The intrinsic dust SED integrates to the absorbed luminosity (to 10^{-5}); the two equilibrium methods (single- T_{eq} Lucy and Bjorkman & Wood) agree on both the absorbed fraction (to $< 0.3\%$ up to $\tau_V = 5$) and the dust temperatures.
- **AMR vs. Cartesian.** Dust emission on a uniform level-5 octree sphere reproduces the reprocessed fraction (0.863 at $\tau_V = 5$) and energy balance of the equivalent Cartesian sphere; Bjorkman & Wood and MRW on the octree conserve energy identically.

External three-dimensional transport benchmarks (DUSTY, Pascucci, the TRUST slab, and a direct Hyperion cross-check) require their published reference solutions and are added as they become available.

9. Algorithm Notes

Packet energy. Each packet carries a physical luminosity L_{pk} [erg/s]: $L_{\star}/N_{\star}$ for stellar packets and $L_{\text{dust}}/N_{\text{dust}}$ for dust-emission packets, so populations with different packet energies combine correctly in the J_{λ} tally.

Lucy iteration. The stellar J_{λ} is tallied once and reused; each iteration transports fresh dust-emission photons (sampled per cell from the emission spectrum), adds their J_{λ} to the stellar field, and recomputes the emission. The total emitted luminosity obeys $L^{(k)} = L_{\text{abs}}^{\star} + f L^{(k-1)}$ with f the reabsorbed fraction, a geometric series converging to $L_{\text{abs}}^{\star}/(1-f)$.

Bjorkman & Wood temperature. A cell's equilibrium temperature is found by inverting the monotonic $\kappa_B(T) = \int \kappa_{\text{abs}}(\lambda) B_\lambda(T) d\lambda$ against the running absorbed power per H nucleus.

References

- Lucy, L. B. 1999, A&A, 344, 282 — iterative mean-intensity / dust-emission estimator.
- Bjorkman, J. E. & Wood, K. 2001, ApJ, 554, 615 — immediate reemission with temperature correction.
- Camps, P. et al. 2015, A&A, 580, A87 — stochastic-heating dust-emission benchmark.
- Zubko, V., Dwek, E., & Arendt, R. G. 2004, ApJS, 152, 211 — BARE-GR-S dust model.
- Min, M. et al. 2009, A&A, 497, 155; Robitaille, T. P. 2010, A&A, 520, A70 — Modified Random Walk.
- Seon, K.-I. 2010, PKAS, 25, 177 — the (a, g) single-run scan.
- Jonsson, P. 2006, MNRAS, 372, 2 — the polychromatic optical-depth scan.
- Górski, K. M. et al. 2005, ApJ, 622, 759 — HEALPix pixelization (all-sky interior observer).
- Amanatides, J. & Woo, A. 1987, Eurographics '87 — fast voxel traversal.